M-DaQ: Retrieving Samples with Multilingual Diversity and Quality for Instruction Fine-Tuning Datasets

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1 MOTIVATION

- Multilingual IFT data is scarce, skewed toward English, lacks systematic curation
- Llama-3 IFT dataset: only 3.01% multilingual samples
- No language-agnostic quality scoring for multilingual data
- Data diversity studied only in English settings
- Superficial Alignment Hypothesis (SAH) unverified in multilingual contexts

Three Challenges Addressed:

C No extensible quality scoring for multilingual IFT
→ QSM with triplet loss

C Diversity selection limited to English
→ DAS inspired by MMR

C SAH unverified in multilingual settings
→ 1K→52K study, 8 langs

2 METHOD: M-DaQ FRAMEWORK

Stage 1: Quality Scoring Model (QSM)

- Fine-tuned on ~2.3K expert-revised samples × 18 languages (from MIDB)
- Triplet loss: instruction → positive (expert-revised) vs negative (original/MT)
- Language-agnostic quality signal from embedding space

Stage 2: Diversity-Aware Selection (DAS)

- MMR-inspired: Quality ≈ Relevance, Diversity ≈ Novelty
- Two-stage pipeline reduces $O(n^2)$ → $O(n \log n)$
- Stage A: Select top-n by QSM score (quality subset)
- Stage B: Greedily augment from uncovered clusters (diversity subset)
- Ratio n_quality : n_diversity = 6:1 (empirically tuned)

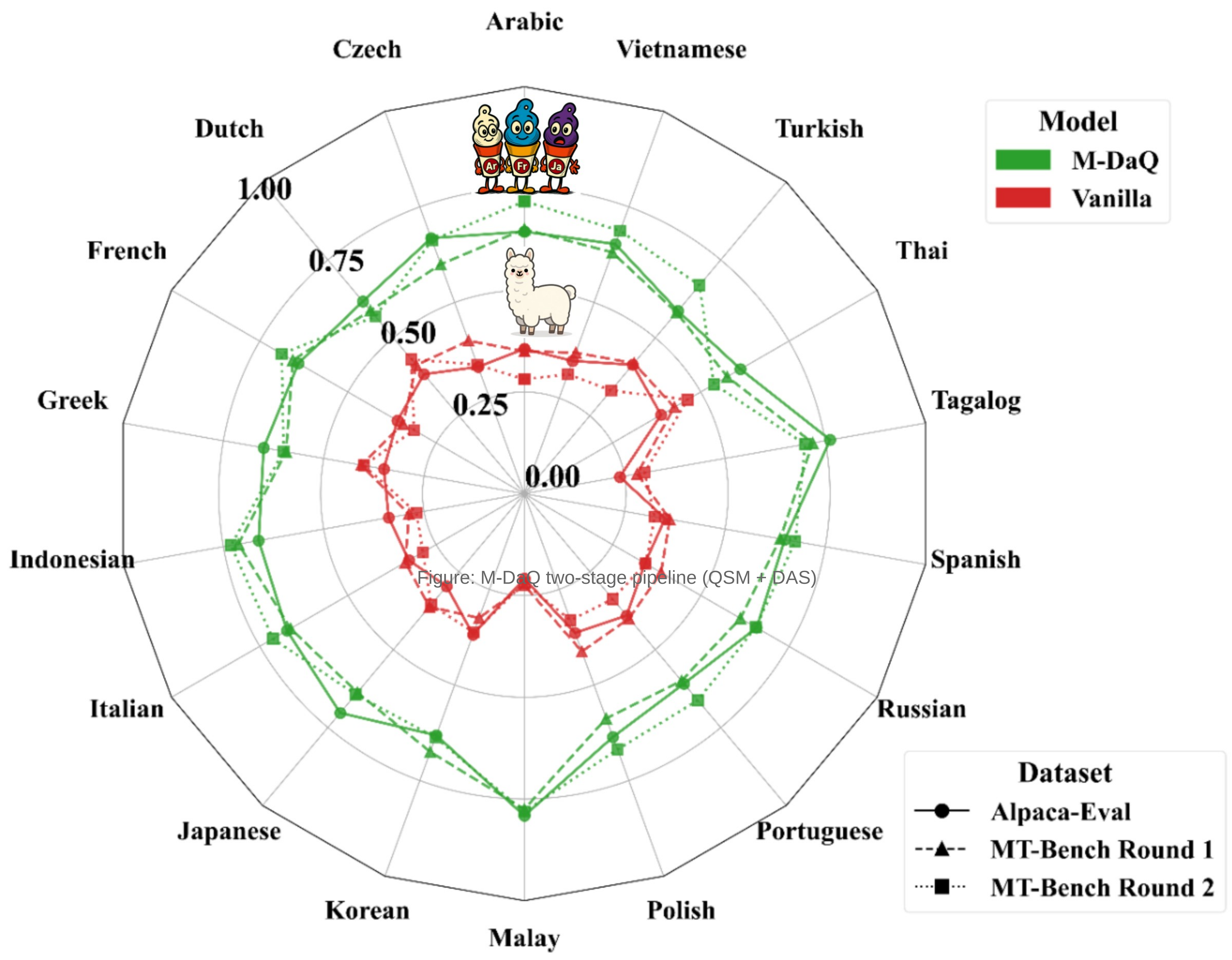
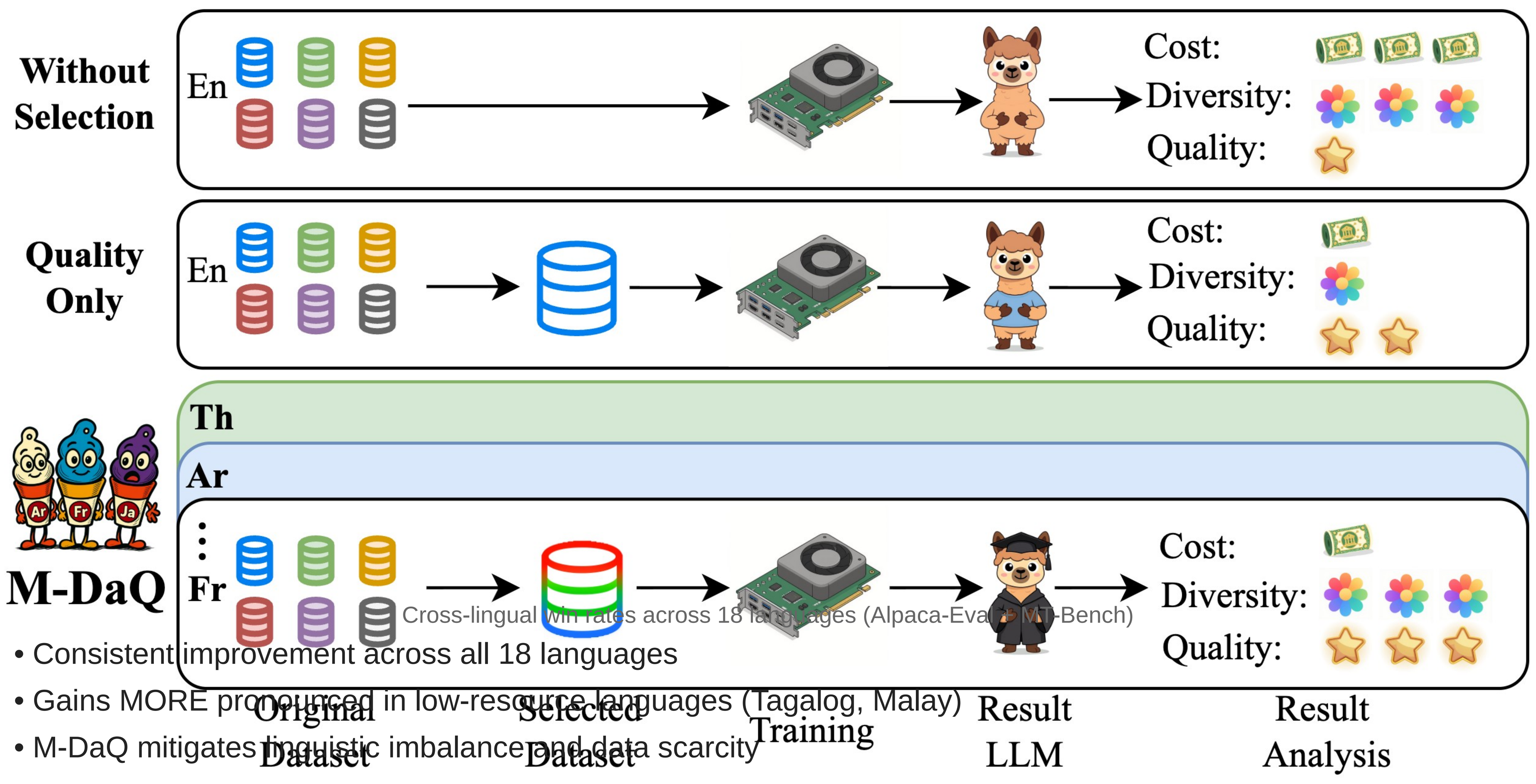


Figure: M-DaQ two-stage pipeline (QSM + DAS)

3 KEY RESULTS

LLM-as-Judge: Avg Win Rate (M-DaQ vs Vanilla)

Alpaca-Eval: 60.2% · MT-Bench R1: 62.6% · MT-Bench R2: 62.9%



Cross-lingual win rates across 18 languages (Alpaca-Eval & MT-Bench)

- Consistent improvement across all 18 languages
- Gains MORE pronounced in low-resource languages (Tagalog, Malay)
- M-DaQ mitigates linguistic imbalance and data scarcity

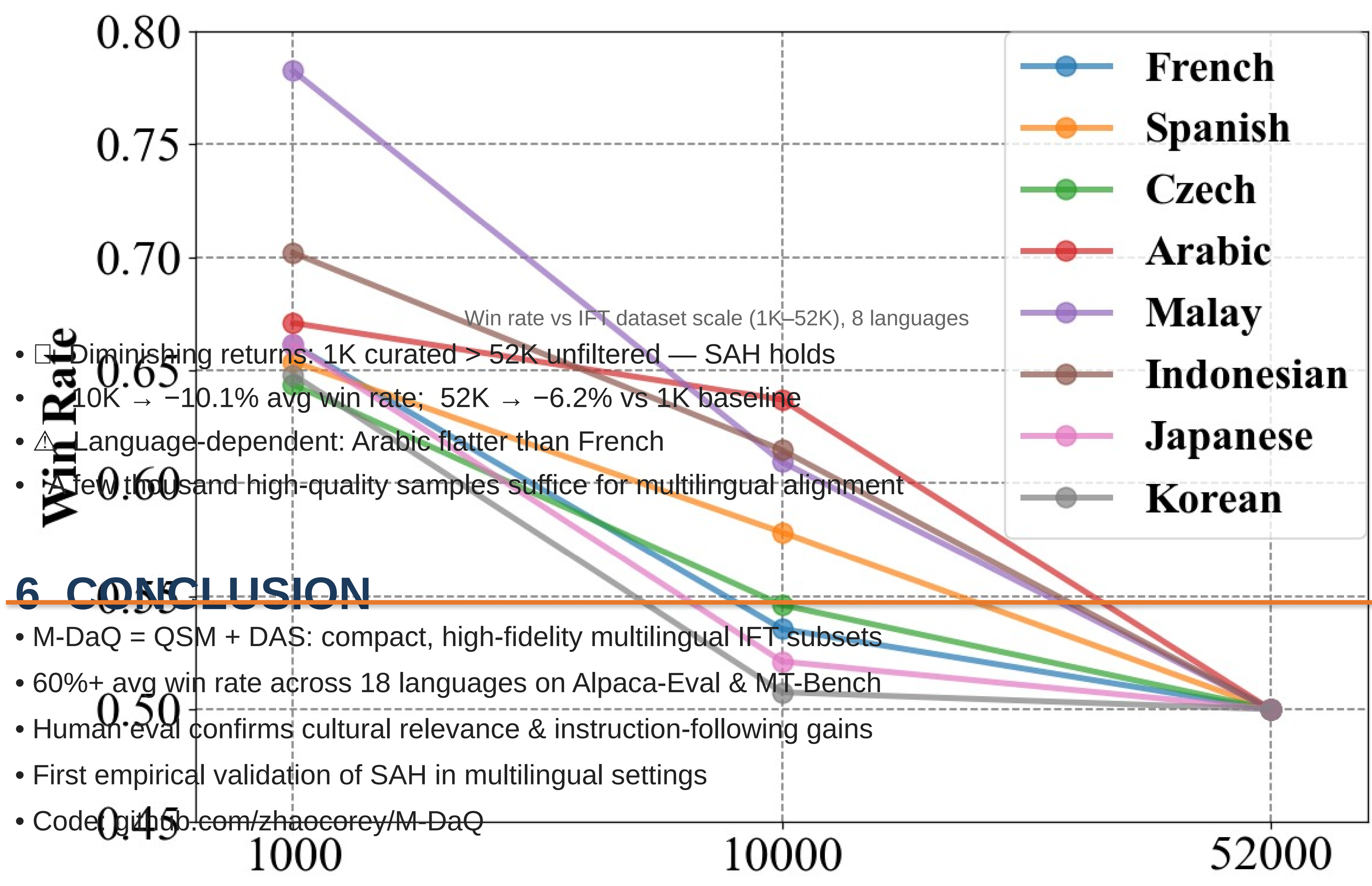
4 HUMAN EVALUATION

6 Languages · 900 Samples · 58 Person-Hours

Language	Alpaca-Eval	MT-Bench R1	MT-Bench R2
Japanese	86%	84%	80%
Korean	94%	94%	94%
Russian	70%	86%	84%
Portuguese	80%	78%	84%
Greek	58%	76%	82%
French	80%	92%	88%
Average	**78.0%**	**85.0%**	**85.3%**

5 SAH IN MULTILINGUAL SETTINGS

First systematic investigation of SAH across languages.



Win rate vs IFT dataset scale (1K–52K), 8 languages

- Diminishing returns: 1K curated > 52K unfiltered — SAH holds
- 10K → −10.1% avg win rate; 52K → −6.2% vs 1K baseline
- ⚠ Language-dependent: Arabic flatter than French
- A few thousand high-quality samples suffice for multilingual alignment

6 CONCLUSION

- M-DaQ = QSM + DAS: compact, high-fidelity multilingual IFT subsets
- 60%+ avg win rate across 18 languages on Alpaca-Eval & MT-Bench
- Human eval confirms cultural relevance & instruction-following gains
- First empirical validation of SAH in multilingual settings
- Code: github.com/zhaocorey/M-DaQ

Code & Resources

github.com/zhaocorey/M-DaQ

Paper

DOI: 10.1145/3805712.3809946 | arXiv: 2509.15549

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